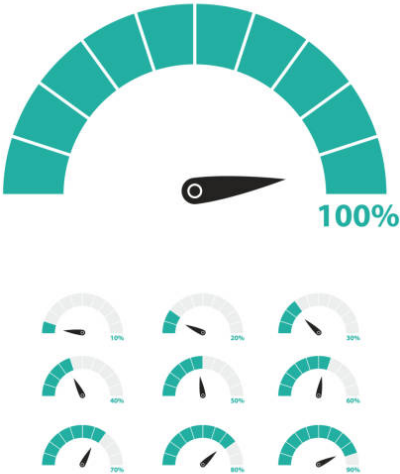


Rating Formula Documentation

Nutritional Products Rater



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1 Objective

We design an objective and reproducible system that, using a supermarket product database, assigns each product a score $[0, 100]$ that reflects two main aspects:

1. **Nutritional quality** (60%): based on an adaptation of the Nutri-Score scheme, explicitly considering energy (kcal), sugars, saturated fats, and salt (sodium), as well as fiber and proteins.
2. **Additives** (40%): penalizes the total number and relative harmfulness of additives, using an exponential function so that the presence of highly harmful additives has a particularly strong impact on the score.

The nutritional component relies on standardized Nutri-Score thresholds, while the additive component is modeled in a parametric but interpretable way. The goal is to maximize *objectivity* (use of existing standards) and *interpretability* (simple and justifiable formulas).

2 Input data

Each row of the product table contains, among others, the fields: `categories_tags`, `source_category`, `additives_n`, `additives_tags`, `energy_kcal_100g`, `sugars_100g`, `saturated_fat_100g`, `fiber_100g`, `proteins_100g`, `salt_100g`.

The fields `categories_tags` and `source_category` are used to identify whether a product is a *beverage* (e.g. if they contain tags such as `beverage`, `drink`, `juice`, `soda`), since Nutri-Score applies stricter energy thresholds and treats proteins differently for beverages.

3 Nutri-Score-based nutritional score

3.1 Negative and positive points

Following the Nutri-Score scheme, we define:

- **Negative** points A (0–40) for:
 - $E = \text{energy_kcal_100g}$
 - $Su = \text{sugars_100g}$
 - $Sa = \text{saturated_fat_100g}$
 - $Na = 400 \cdot \text{salt_100g}$ (sodium in mg)
- **Positive** points C (0–10) for:
 - $Fi = \text{fiber_100g}$
 - $Pr = \text{proteins_100g}$

Each magnitude is converted into points using piecewise functions $a_E, a_{Su}, a_{Sa}, a_{Na}, c_{Fi}, c_{Pr}$, with thresholds defined in the Nutri-Score literature. For example, for solid foods:

- Energy E (kcal/100g): 0 points if $E < 50$, 1 point if $50 \leq E < 80$, ..., 10 points if $E \geq 450$.
- Sugars Su (g/100g): 0 points if $Su < 4.5$, 1 point if $4.5 \leq Su < 9$, ..., 10 points if $Su \geq 45$.

- Saturated fats Sa (g/100g): 0 points if $Sa < 1$, 1 point if $1 \leq Sa < 2$, ..., 10 points if $Sa \geq 10$.
- Sodium Na (mg/100g): 0 points if $Na < 90$, 1 point if $90 \leq Na < 180$, ..., 10 points if $Na \geq 900$.

For beverages, energy thresholds are stricter (e.g. 0 points if $E < 10$, 1 point if $10 \leq E < 20$, etc.), and protein is usually not rewarded.

For positive points:

- Fiber Fi (g/100g): 0 points if $Fi < 1$, 1 point if $1 \leq Fi < 2$, ..., 5 points if $Fi \geq 4$.
- Protein Pr (g/100g): 0 points if $Pr < 1.6$, 1 point if $1.6 \leq Pr < 3.2$, ..., 5 points if $Pr \geq 8$. For beverages, protein is not considered in the positive score.

We group negative and positive points as:

$$A = a_E(E) + a_{Su}(Su) + a_{Sa}(Sa) + a_{Na}(Na), \quad A \in [0, 40],$$

$$C = c_{Fi}(Fi) + c_{Pr}(Pr), \quad C \in [0, 10].$$

3.2 Raw score and normalization

The raw Nutri-Score is defined as:

$$R = A - C,$$

where R typically takes values in $[-10, 40]$, with lower values indicating better nutritional quality.

To obtain a nutritional score in the range $[0, 100]$, we apply a linear transformation:

$$Q_{\text{nutri}} = \text{clip}_{[0,100]} \left(100 \cdot \frac{40 - R}{50} \right)$$

Thus, approximately:

- If $R = -10$ (excellent nutritional profile), $Q_{\text{nutri}} \approx 100$.
- If $R = 40$ (very poor nutritional profile), $Q_{\text{nutri}} \approx 0$.

This construction explicitly and objectively incorporates:

- The product's energy (kcal).
- Sugar content in *all* products (not just beverages).
- Saturated fats and salt/sodium.
- Fiber and proteins as mitigating factors (positive points).

4 Formula for Q_{add} (additives)

Let $\mathcal{E}$ be the set of additives present in the product. We assume we have (optionally) a dictionary D assigning each additive $e \in \mathcal{E}$ a hazard category:

$$D(e) \in \{\text{non_harmful}, \text{harmful}, \text{very_harmful}\}.$$

We define a numerical severity $s(e)$ as:

$$s(e) = \begin{cases} 0, & \text{if } D(e) = \text{non_harmful}, \\ 1, & \text{if } D(e) = \text{harmful}, \\ 3, & \text{if } D(e) = \text{very_harmful}. \end{cases}$$

The total severity of the product is then:

$$S = \sum_{e \in \mathcal{E}} s(e).$$

If no dictionary is provided (default development configuration), we simply take $s(e) = 1$ for all additives, i.e. S matches the total number of declared additives.

Instead of linear penalization, we use an **exponential penalty** on severity:

$$Q_{\text{add}} = \text{clip}_{[0,100]}(100 \cdot \exp(-\lambda S))$$

where $\lambda > 0$ is a calibration parameter. In our experiments we used $\lambda \approx 0.25$.

Justification of the exponential form. The exponential function has two key advantages:

- A single *very harmful* additive ($s = 3$) already produces a noticeable drop in Q_{add} (for example, with $\lambda = 0.25$, $S = 3$ yields $Q_{\text{add}} \approx 47$), which visibly reduces the final score.
- The penalty becomes *more severe* as the number of harmful additives increases, while keeping an easy interpretation in terms of total severity S .

5 Formula for the final score

The total score is computed as a weighted average of both components:

$$\text{health_score} = \text{round}(0.60 \cdot Q_{\text{nutri}} + 0.40 \cdot Q_{\text{add}})$$

Thus, nutritional quality (based on Nutri-Score) is the main determinant of the score, while the presence of additives (especially very harmful ones) can substantially reduce the final rating.

6 Reasoning and design decisions

- **Based on existing standards:** the nutritional part relies on the points and thresholds of the *Nutri-Score* system, which objectively quantifies the impact of energy, sugars, saturated fats, and salt/sodium, mitigated by fiber and proteins.
- **Sugar in all products:** sugar is penalized in every category, not only beverages, consistent with the Nutri-Score approach. In beverages, only the energy thresholds change.
- **Additives with non-linear impact:** by using an exponential function in Q_{add} , the presence of very harmful additives can significantly reduce the score even if the nutritional profile is good.
- **Modularity:** the severity $s(e)$ can be refined as better evidence for each additive becomes available. The mathematical framework (sum of severities followed by an exponential) remains unchanged.
- **Computational simplicity:** all operations are local to each product (sum of points and elementary functions), so the algorithm scales well to large datasets.

7 Summary of the processing flow in code

1. **Detect** whether the product is a beverage using `categories_tags/source_category`.
2. **Compute** negative points A (energy, sugars, saturated fats, sodium) and positive points C (fiber, proteins) using Nutri-Score tables.
3. **Obtain** the raw score $R = A - C$ and normalize it to $[0, 100]$ to obtain Q_{nutri} .
4. **Compute** the total additive severity S (either from a dictionary D or assuming unit severity).
5. **Compute** $Q_{\text{add}} = \text{clip}_{[0,100]}(100 \exp(-\lambda S))$.
6. **Combine** both components with $\text{health_score} = \text{round}(0.60 Q_{\text{nutri}} + 0.40 Q_{\text{add}})$.

8 Practical considerations and calibration

- **Soft drinks and sugary beverages:** tend to accumulate many negative points for sugars and energy, with stricter thresholds in beverages, resulting in low Q_{nutri} values.
- **100% fruit juices:** although they may have good fiber profiles, their sugar content is reflected in the nutritional score. If a more permissive treatment is desired, sugar or energy thresholds for beverages can be adjusted slightly.
- **Cheeses and oils:** penalized for energy and saturated fats, but may partially benefit from good protein content (cheeses) or other criteria if the model is extended. This is an inherent limitation of Nutri-Score.
- **Very harmful additives:** the parameter λ controls severity. Larger values make a few very harmful additives significantly drop Q_{add} ; smaller values make the system more tolerant.

9 Nutri-Score cutoff tables used

Below is a summary of the nutritional ranges used to compute negative and positive points in Nutri-Score. These values were extracted from the original Nutri-Score technical document, OpenFoodFacts technical documentation, and the evidence summary published by IARC/WHO.

Negative points A (0–40)

Negative points come from four nutrients: energy, sugars, saturated fats, and sodium. Each is scored from 0 to 10 according to official ranges. In our dataset, sodium is obtained as:

$$\text{sodium (mg)} = 400 \cdot \text{salt (g)}.$$

Points	0	1	2	3	4	5	6	7	8	9	10
Energy (kcal/100g)	< 50	50–80	80–110	110–140	140–170	170–200	200–230	230–270	270–310	310–450	≥ 450
Sugars (g/100g)	< 4.5	4.5–9	9–13.5	13.5–18	18–22.5	22.5–27	27–31	31–36	36–40	40–45	≥ 45
Saturated fats (g/100g)	< 1	1–2	2–3	3–4	4–5	5–6	6–7	7–8	8–9	9–10	≥ 10
Sodium (mg/100g)	< 90	90–180	180–270	270–360	360–450	450–540	540–630	630–720	720–810	810–900	≥ 900

For beverages, only energy thresholds change, becoming stricter (e.g. 0 points if $E < 10$, 1 point if $10 \leq E < 20$, etc.).

Energy thresholds comparison

Points	0	1	2	3	4	5	6	7	8	9	10
Food (kcal/100g)	< 50	50–80	80–110	110–140	140–170	170–200	200–230	230–270	270–310	310–450	≥ 450
Beverages (kcal/100ml)	< 10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100	≥ 100

Positive points C (0–10)

Fiber and proteins contribute up to 5 points each (proteins are not rewarded in beverages). The official cutoffs are:

Fiber points	0	1	2	3	4	5
Fiber (g/100g)	< 1	1–2	2–2.8	2.8–3.4	3.4–3.9	≥ 3.9

Protein points	0	1	2	3	4	5
Proteins (g/100g)	< 1.6	1.6–3.2	3.2–4.8	4.8–6.4	6.4–8	≥ 8

Once negative points A and positive points C are obtained, the raw Nutri-Score value is:

$$R = A - C,$$

which is then rescaled to the continuous $[0, 100]$ scale via:

$$Q_{\text{nutri}} = \text{clip}_{[0,100]} \left(100 \cdot \frac{40 - R}{50} \right).$$

10 Sources of the tables used

The above ranges come from three official and verified sources:

- **OpenFoodFacts – Nutri-Score algorithm documentation and implementation**
Reproducible implementation of the algorithm and technical description of nutritional cutoffs. <https://openfoodfacts.github.io/openfoodfacts-server/api/>
- **Official Nutri-Score scientific document (Santé Publique France)** Locatable via: [site:santepubliquefrance.fr "scientific and technical report" "nutri-score"](https://www.santepubliquefrance.fr/scientific-and-technical-report/nutri-score)
Includes the full tables of negative and positive points.
- **WHO / IARC – Evidence Summary Brief on Nutri-Score** Scientific summary of how the Nutri-Score system works. https://www.iarc.who.int/wp-content/uploads/2021/09/IARC_Evidence_Summary_Brief_2.pdf

These sources allow faithful reconstruction of the nutritional component of Nutri-Score and ensure that the ranges used in this project are consistent with the official standard.